USN | | | | | | | | | | [22CSE368]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

PRINCIPLES OF VIRTUALIZATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	How would you implement virtualization in a data center to improve resource allocation and management?	L3	CO1	**02**
	b.	Given a set of privileged instructions, how would you use binary translation in a virtualized environment to enhance performance?	L3	CO2	**02**
	c.	Analyze the complexities of memory virtualization in a large-scale cloud environment, and how would you propose solutions.	L4	CO3	**02**
	d.	Evaluate how would you assess the effectiveness of high availability and fault tolerance mechanisms in minimizing downtime for mission-critical applications.	L5	CO3	**02**
	e.	Analyze the role of the VMware Platform Service Controller in ensuring secure authentication and license management across virtualized environments.	L4	CO3	**02**
		PART-B			**6 X 5=30**
2.	**a.**	In a scenario with large storage demands, how would you apply storage virtualization techniques to improve storage management and scalability?	L3	CO2	**05**
		OR			
	b.	Analyze the benefits and challenges of memory and I/O virtualization in a system that requires high data throughput.	L4	CO2	**05**
3.	**a.**	Analyze how you would compare Type 1 and Type 2 hypervisors in terms of performance, security, and resource management, and decide which is more suitable for enterprise virtualization.	L4	CO2	**05**
		OR			
	b.	Evaluating the performance trade-offs between para-virtualization and full virtualization in a high-performance application environment.	L5	CO3	**05**
4.	**a.**	Apply IP addressing schemes in a virtualized network to ensure efficient communication between virtual machines across multiple subnets.	L3	CO3	**05**
		OR			

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	b.	How would you apply RAID configurations in a virtualized environment to ensure data redundancy and fault tolerance?	L3	CO3	**05**
5.	**a.**	Analyze the key management challenges in a data center with a complex virtualized environment, and what strategies would you propose to mitigate them.	L4	CO4	**05**
		OR			
	b.	Design a data center solution that integrates load balancing, DRS, and DPM to optimize both performance and energy consumption.	L6	CO4	**05**
6.	**a.**	Develop a plan to handle the scalability challenges in a data center that needs to expand its resources while maintaining high availability.	L6	CO4	**05**
		OR			
	b.	Define Virtual LAN (VLAN) and its role in network segmentation.	L1	CO3	**05**
7.	**a.**	How would you apply memory paging to optimize virtual machine performance in a resource-constrained environment?	L3	CO3	**05**
		OR			
	b.	How would you apply storage area network (SAN) to improve storage scalability and performance in a cloud environment?	L3	CO3	**05**
		PART-C			**6 X 10=60**
8.	**a.**	Create a virtualization infrastructure using both Type 1 and Type 2 hypervisors to manage workloads in a hybrid cloud environment.	L6	CO2	**10**
		OR			
	b.	Design a server consolidation strategy using virtualization technologies to reduce physical server count while maintaining high availability.	L6	CO2	**10**
9.	**a.**	Analyze the impact of RAID levels (RAID 0, 1, 5, etc.) on performance and data redundancy in a virtualized storage system.	L4	CO3	**10**
		OR			
	b.	Compare the performance and memory management efficiency of virtual memory versus physical memory in virtualized environments.	L4	CO3	**10**
10.	**a.**	How would you apply fault tolerance to ensure the availability of critical applications in a virtualized environment?	L3	CO4	**10**
		OR			

	b.	How would you implement load balancing in a cloud infrastructure to distribute traffic efficiently across virtual machines?	L3	CO4	**10**
11.	**a.**	Evaluate the scalability and performance improvements offered by VMware VSAN in a growing virtualized environment. Would you recommend it over traditional storage solutions?	L5	CO5	**10**
		OR			
	b.	Evaluate the impact of enabling Port Mirroring in a virtualized environment on network monitoring and security. Is it a viable solution for real-time network traffic analysis?	L5	CO5	**10**
12.	**a.**	Construct a system that uses live migration, vMotion, and Storage vMotion to ensure seamless maintenance and resource allocation in a high-demand environment.	L6	CO5	**10**
		OR			
	b.	Design a virtualized infrastructure for a large enterprise using VMware vSphere, integrating VSAN, vCenter Server, and a Virtual Distributed Switch for optimal performance and scalability.	L6	CO5	**10**
13.	**a.**	Create a plan to implement LACP and NetFlow in a virtualized environment using a Virtual Distributed Switch to enhance network throughput and monitoring capabilities.	L6	CO5	**10**
		OR			
	b.	Construct a system that leverages both VMware vSphere and Microsoft Hyper-V to handle multi-platform virtual machine management within a hybrid cloud environment.	L6	CO5	**10**